

PyPSA-Eur — Industrial Energy-Demand Configuration Assumptions

All keys are under the top-level **industry**: block unless prefixed **sector**. Year-indexed values are interpolated per planning horizon; endpoints shown as 2020 to 2050. Defaults from `config/config.default.yaml`.

Per-node energy demand = sum over subsectors of [(A) route shares × (B) baseline production, giving tonnes] × [(C/D) intensities blended by (E) toward the future], with (F)/(G) controlling what is included and how it is placed.

A. Production route shares — what technology makes each material

Used in `build_industrial_production_per_country_tomorrow.py` to split tonnage across routes.

Key	Default (2020–2050)	Unit	Assumption
St_primary_fraction	0.60 to 0.30	fraction	Steel via primary (BF/DRI) vs. secondary (scrap+EAF)
DRI_fraction	0 to 1.0	fraction	Share of <i>primary</i> steel via H ₂ -DRI + electric arc
Al_primary_fraction	0.40 to 0.20	fraction	Aluminium via primary (electrolysis) vs. secondary (scrap)
HVC_primary_fraction	0.88 to 0.40	fraction	High-value chemicals from virgin feedstock
HVC_mechanical_recycling_fraction	0.12 to 0.30	fraction	HVC from mechanical recycling
HVC_chemical_recycling_fraction	0 to 0.20	fraction	HVC from chemical recycling
HVC_environment_sequestration_fraction	0.0	fraction	HVC landfilled (carbon sink) rather than combusted

B. Reference-year baseline production — denominators for splitting chemicals

Used in `build_industrial_production_per_country.py` and to normalise chemical sub-demands.

Key	Default	Unit	Assumption
basic_chemicals_without_NH3_production_today	69.0	Mt/a	Basic chemicals excl. ammonia (= 86 - 17 Mt)
HVC_production_today	52.0	Mt/a	High-value chemicals output
chlorine_production_today	9.58	Mt/a	Chlorine output
methanol_production_today	1.5	Mt/a	Methanol output

C. Technology energy intensities — MWh per tonne of material

Used in `build_industry_sector_ratios.py` (and the ammonia/chlorine/methanol split in `build_industrial_energy_demand_per_country_today.py`).

Key	Default	Unit	Assumption
H2_DRI	1.7	MWh _{H₂} / t steel	Hydrogen for DRI
elec_DRI	0.322	MWh _e / t steel	Electricity for DRI shaft

Key	Default	Unit	Assumption
MWh_NH3_per_tNH3	5.166	MWh / t NH ₃	Energy content of ammonia (LHV)
MWh_CH4_per_tNH3_SMR	10.8	MWh _{CH4} / t NH ₃	Methane for ammonia via SMR
MWh_elec_per_tNH3_SMR	0.7	MWh _e / t NH ₃	Electricity for ammonia via SMR
MWh_H2_per_tNH3_electrolysis	5.93	MWh _{H2} / t NH ₃	Hydrogen for electrolytic Haber-Bosch
MWh_elec_per_tNH3_electrolysis	0.2473	MWh _e / t NH ₃	Electricity for electrolytic Haber-Bosch
MWh_NH3_per_MWh_H2_cracker	1.46	MWh _{NH3} / MWh _{H2}	Ammonia consumed per H ₂ from cracking
MWh_elec_per_tHVC_mechanical_recycling	0.547	MWh _e / t HVC	Electricity for mechanical recycling
MWh_elec_per_tHVC_chemical_recycling	6.9	MWh _e / t HVC	Electricity for chemical recycling
MWh_elec_per_tCl	3.6	MWh _e / t Cl	Electricity for chlorine (chloralkali)
MWh_H2_per_tCl	-0.9372	MWh _{H2} / t Cl	Hydrogen co-produced with chlorine (negative)
MWh_elec_per_tMeOH	0.167	MWh _e / t MeOH	Electricity for methanol from gas
MWh_CH4_per_tMeOH	10.25	MWh _{CH4} / t MeOH	Methane for methanol from gas
MWh_MeOH_per_tMeOH	5.528	MWh / t MeOH	Energy content of methanol (LHV)

D. Process-emission assumptions — CO₂ not from fuel combustion

Used in build_industry_sector_ratios.py to attribute the 'process emission' / 'process emission from feedstock' rows.

Key	Default	Unit	Assumption
NH3_process_emissions	24.5	MtCO ₂ /a (EU ref.)	Ammonia-SMR process CO ₂ , netted out of basic chemicals
petrochemical_process_emissions	25.5	MtCO ₂ /a (EU ref.)	HVC feedstock process CO ₂ , moved to the feedstock row
oil_refining_emissions	0.013	tCO ₂ / MWh	CO ₂ from oil fuel processing (refining)

E. Efficiency transition & reference year — how the intensities evolve

Key	Default (2020–2050)	Unit	Assumption
sector_ratios_fraction_future	0.0 to 1.0	fraction	Blend of today's vs. best-in-class specific energy consumption (build_industry_sector_ratios_intermediate.py)
reference_year	2023	year	Baseline year for JRC-IDEES energy balances & production

F. Options, spatial data & removal

Key	Default	Unit	Assumption
hotmaps_locate_missing	false	bool	Geocode industrial sites lacking coordinates (build_industrial_distribution_key.py)

Key	Default	Unit	Assumption
waste_to_energy	false	bool	Enable waste-to-energy CHPs (plastics)
waste_to_energy_cc	false	bool	... with carbon capture
endogenous_subsectors	[]	list	(new) Subsectors dropped from the exogenous energy demand because modeled downstream (e.g. MacroEnergy.jl)

G. sector: toggles that also shape industrial demand

Key	Default	Effect
sector.industry	true	Master switch for the industry module
sector.ammonia	true	If truthy, ammonia is a separate endogenous carrier; else decomposed into H ₂ + electricity in the demand
sector.HVC_demand_factor	1.0	Scales HVC (naphtha) demand — applied in add_industry at network-build time